

nf-core/funcprofiler

Standardised functional profiling of metagenomes at any scale



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Motivation

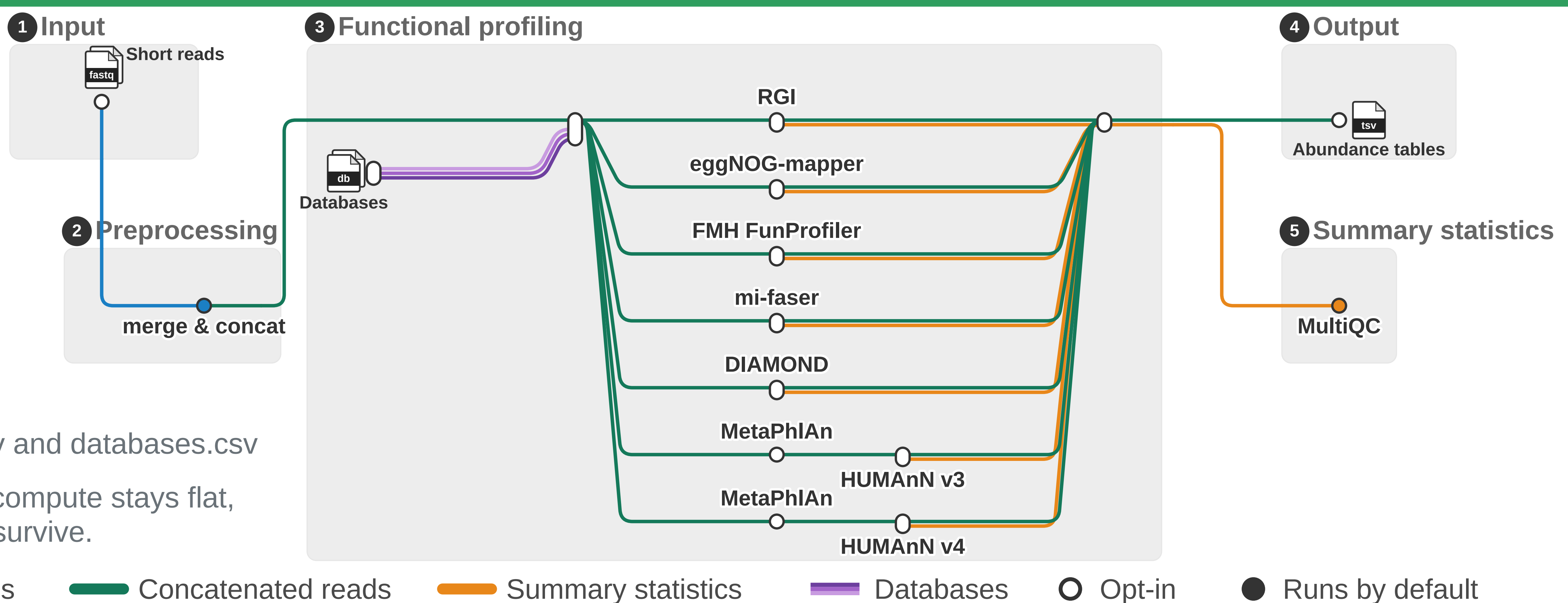
The problem

- Tools, databases and ontologies are fragmented
- Results rarely compare across studies
- Taxonomy has community workflows; function does not

What a community workflow must give

- Many profilers in one pass, from one input
- Native outputs kept, no forced harmonisation
- Containers, continuous testing, versioned releases

Seven profilers, one samplesheet



References

HUMANv · 10.7554/eLife.65088 · FMH FunProfiler · 10.1093/bioinformatics/btae397 · RGI / CARD · 10.1093/nar/gkac920
mifaser · 10.1093/nar/gkx1209 · DIAMOND · 10.1038/s41592-021-01101-x · eggNOG-mapper · 10.1093/molbev/msab293

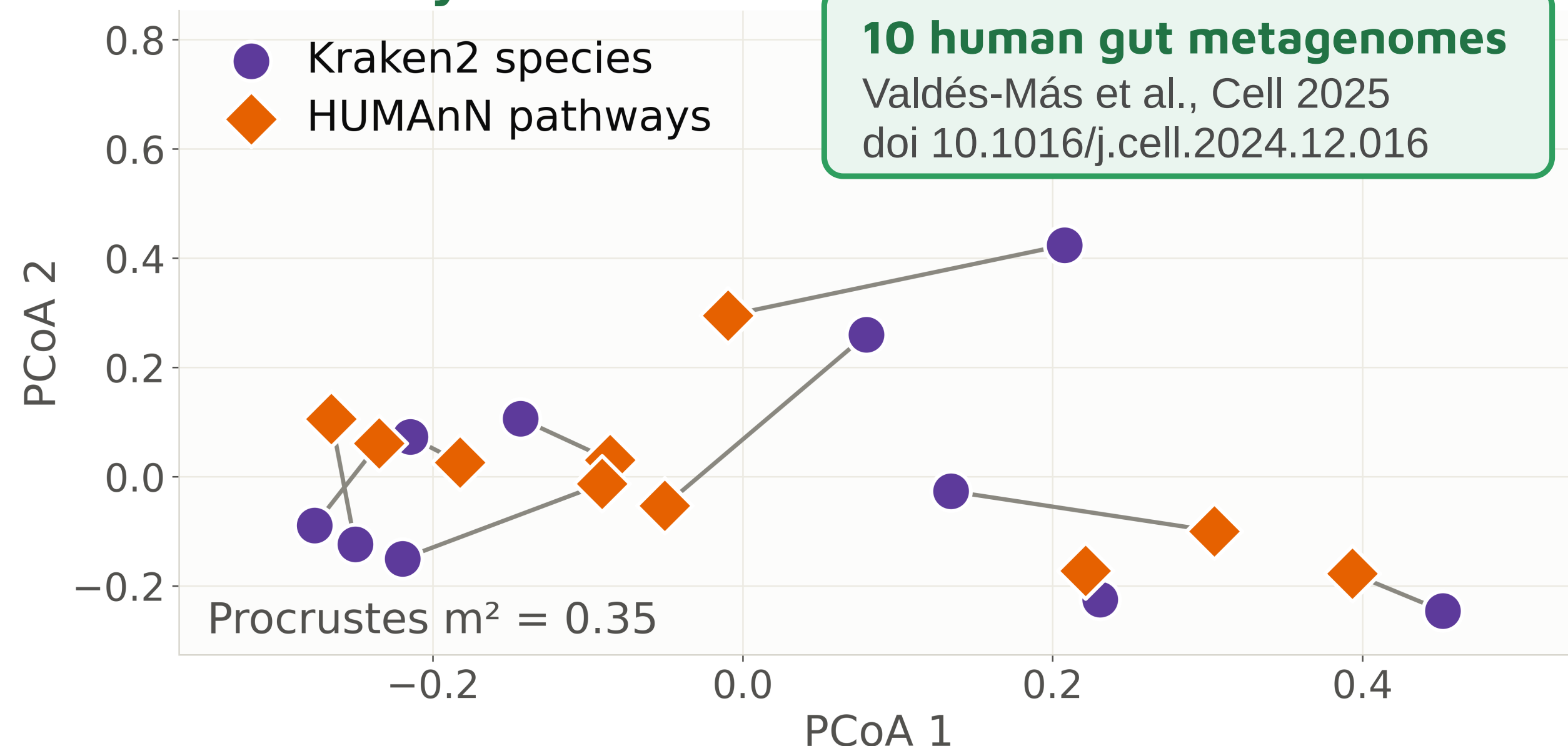


Px4 Health: Personalised, Precise, Predictive, Preventative. A prospective study of 10,000+ healthy Australian adults, pairing lifestyle and clinical phenotyping with genomes, proteomes, metabolomes and lipidomes from blood, saliva, urine and stool. The “Our Health Study” has adopted funcprofiler.

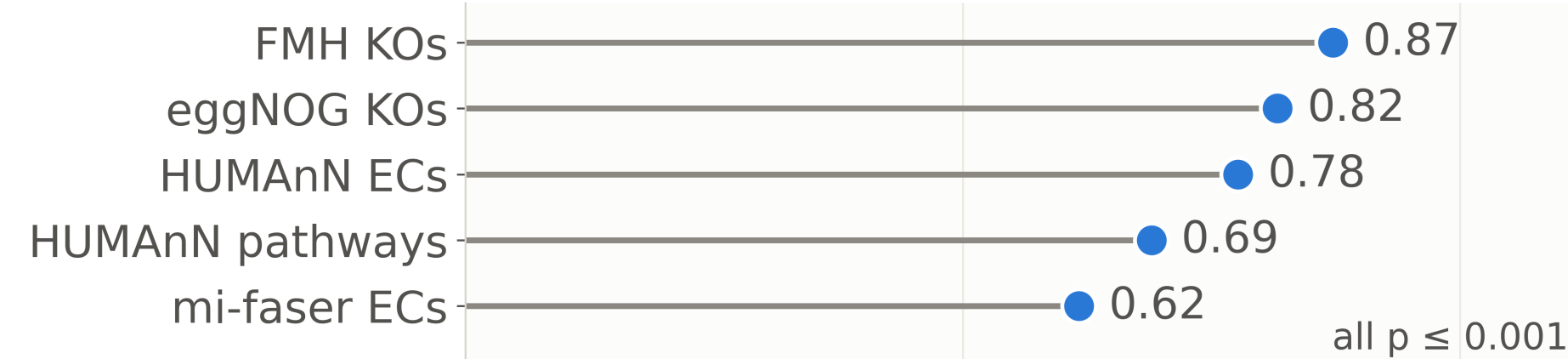
px4health.org

Applications

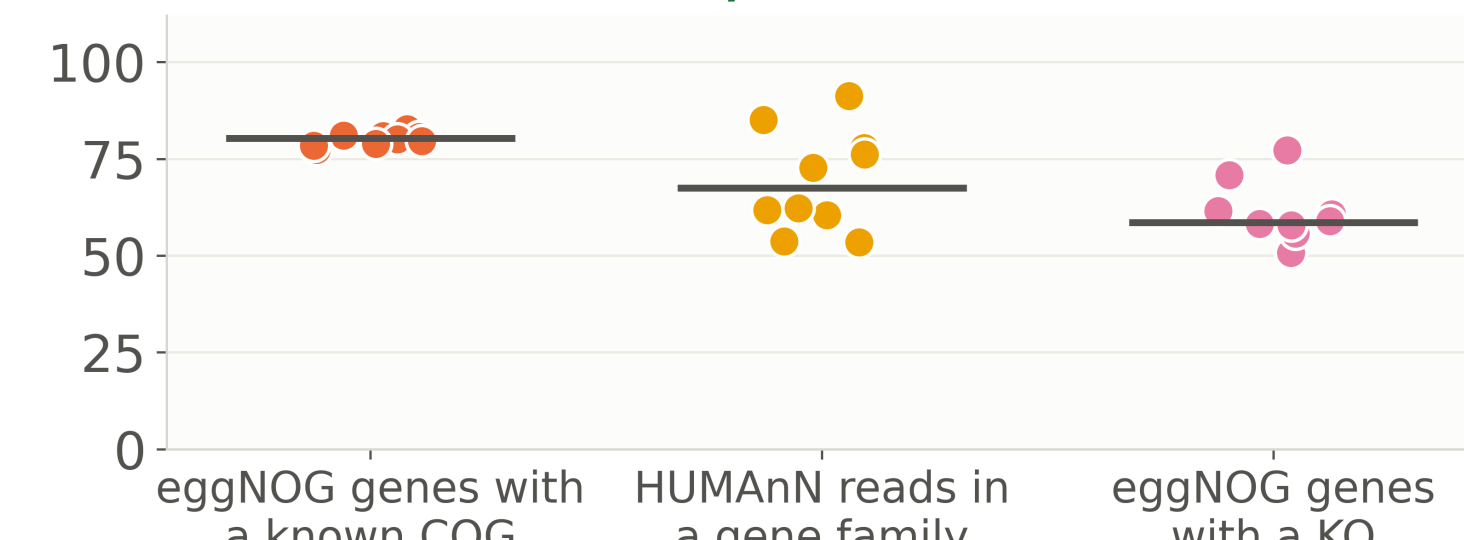
Taxonomy vs function



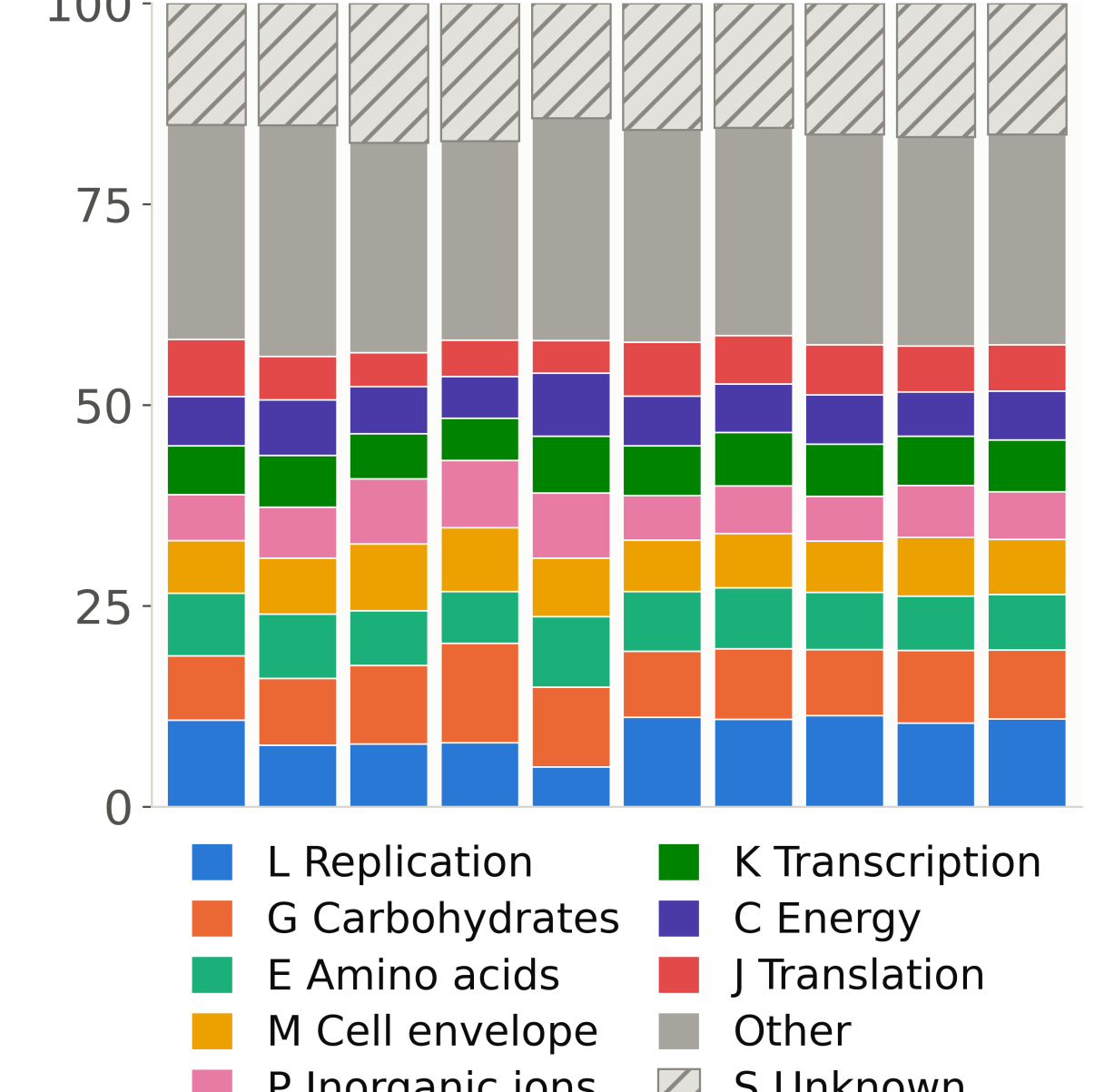
Mantel r vs Kraken2 species



Given a function, % annotated



eggNOG COG categories, %



Future

PRIORITY

- Long-read support.** Nanopore and PacBio through the same samplesheet, with long-read QC and assembly-free profiling.
- Consensus profiles: one call set from several tools.
- A companion workflow to build and version profiler databases.

Get involved!

nf-co.re/funcprofiler

nf-co.re/join

#funcprofiler

nextflow run nf-core/funcprofiler

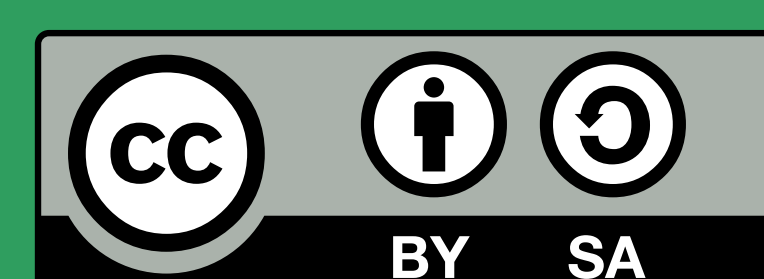
Development team



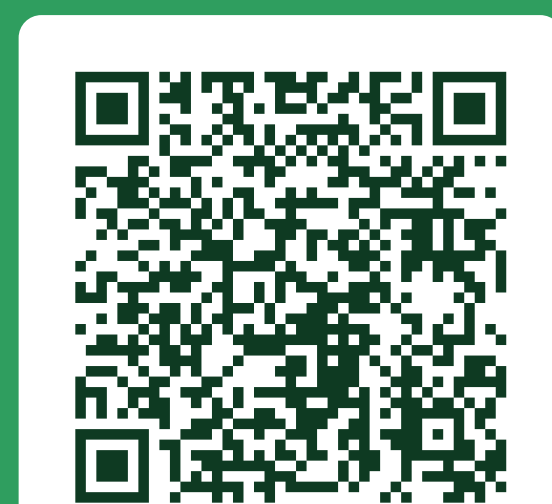
@nickp60 @miraep8 @Yixuan39



Supported by Melbourne Bioinformatics and Px4 Health (MRFF NCRI000043).



Icons: openmoji.org



poster



code